

EchoBraid-Health × INTRINSICA — Enhanced UI Blueprint v1.1

A neurodivergent-aligned, consent-first interface—now mapped to INTRINSICA's data/AI plane, governance, and interop layers for production-grade delivery.

A) What's New in v1.1 (at a glance)

- **Five-layer mapping** of your Experience Stack to INTRINSICA: *Experience* → *Services* → *Data/AI* → *Interop* → *Governance*.
 - **Consent-first data contracts** (FHIR-aligned) for meals, hydration, sleep, and signals; **purpose-bound sharing** and **fail-closed** behaviors.
 - **Resonance safety loop** operationalized: $\Delta\Lambda^p$ drift + Forecast Entropy Gradient (FEG) → *CSL gate* → *Omega Action* (Sip, Salt, Snack, Step, Still).
 - **Offline-first** patterns with 24h sync tolerance and **sovereign nodes** for clinics/coaches.
 - **Pilot-ready scope**: stitches to Pre-hybrid MVP (90-day pilot), with a runway to MVP-2 (full precision health stack).
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B) Stack Mapping to INTRINSICA

1) Experience Layer (EchoBraid-Health UI)

- **Home tiles** (nonlinear): Quick Start, Log Meal, Hydration, Reflect, Plan, Coherence Map, Groups, Enter in Silence.
- **Session—Soft Loop**: Dialogue pane, Breath toggle, Offer Silence, Tone meter, Gentle Prompt Picker.
- **Reflect—Holo-Scriptor**: Prime Prompt Board, Spiral Timeline, tag feelings/needs/symptoms, private notes.
- **Coherence (Maps)**: Resonance Dashboard with Energy/Hydration/Meal Rhythm/Sleep/ $\Delta\Lambda^p$ /FEG; Omega Actions.
- **Plan & Meals**: Micro-protocols (n-of-1) + simple meal log; no calorie requirements.
- **Groups (Microcircle)**: Consent-gated pulse sessions; Alignment Check before any sharing.

2) Services Layer (new orchestration micro-services)

- **Prompt Orchestrator**: delivers *gentle, reversible* micro-prompts; tags each with purpose + consent scope.
- **Rhythm Builder**: composes and schedules micro-protocols; enforces “one change per 1–6 weeks”.
- **Omega Action Engine**: chooses *Sip/Salt/Snack/Step/Still* based on $\Delta\Lambda^p$ /FEG + user preferences (e.g., “warm water”).
- **CSL Gate (Cognitive Strain Limiter)**: limits prompt frequency, length, and intensity; falls back to Silence Mode.

- **Reciprocity Balancer:** guarantees the UI “asks less than it gives” (copy, summaries, auto-logs) to reduce effort.
- **Report Composer:** one-tap opt-in exports for coach/clinician (de-identifiable presets + redaction).

3) Data/AI Layer (INTRINSICA)

- **Signals & features:** satiety trend, hydration sufficiency, meal rhythm, sleep regularity, drift risk ($\Delta\Lambda^p$), FEG trend.
- **Explainability:** every suggestion stamped with inputs → features → rule/operators → Omega Action.
- **Risk operators:** Prime-indexed, smoothness-certified operators govern drift detection; outputs bounded to non-coercive supports.

4) Interop Layer

- **FHIR R4 resources** (examples below); **xAPI** for reflection learning events; **DID-backed IDs** for shared nodes.
- **One-tap export:** patient-controlled bundles (JSON, PDF) with consent ledger stamps.

5) Governance Layer

- **Consent & purpose binding:** every data flow carries a purpose tag; default **24h auto-erase** for session text.
- **Ethical firewalls:** diet-punitive language filters; trigger-aware copy; human escalation only with explicit consent.
- **Equity & ND metrics:** overload rate, prompt debt, access/readability; audit log is user-visible.

C) Data Contracts (minimal, auditable)

1) Core Entities (conceptual)

- **User, Session, Entry** (meal/hydration/movement/sleep/symptom/mood), **Protocol, Metric, ConsentGrant**.
- **Privacy:** local IDs by default; export = explicit; redaction fields on all text.

2) Event Schemas (JSON examples)

Entry: Meal (local, FHIR-mappable)

```
{
  "id": "loc-evt-9f23",
  "type": "meal",
  "ts": "2025-12-22T14:12:03Z",
  "fields": {
    "mealType": "breakfast",
    "goToId": "eggs-yogurt-banana",
  }
}
```

```

    "satietyTag": "steady",
    "context": ["budget", "time-pressed"],
    "notesPrivate": "warm foods preferred"
  },
  "consent": {"share": false, "purpose": "self"},
  "mappings": {
    "FHIR": [
      {"resource": "Observation", "profile": "NutritionIntake", "loinc":
"96248-6"}
    ]
  }
}

```

Entry: Hydration

```

{
  "id": "loc-evt-b101",
  "type": "hydration",
  "ts": "2025-12-22T15:03:01Z",
  "fields": {"size_oz": 10, "tempPref": "warm", "sodiumHint": true},
  "consent": {"share": false, "purpose": "self"},
  "mappings": {"FHIR": [{"resource": "Observation", "loinc": "29274-8"}]}
}

```

Entry: Sleep (summary)

```

{
  "id": "loc-evt-slp77",
  "type": "sleep",
  "ts": "2025-12-22T11:10:00Z",
  "fields": {"duration_min": 420, "efficiency": 0.83, "latency_min": 22},
  "source": {"wearable": "ring", "confidence": 0.82},
  "consent": {"share": true, "scope": "coach"},
  "mappings": {"FHIR": [{"resource": "Observation", "loinc": "93832-4"}]}
}

```

ConsentGrant

```

{
  "id": "cons-47a2",
  "subject": "user:local-001",
  "grantee": "coach:abc123",
  "scope": ["sleep", "protocol"],

```

```
"purpose": "care-coaching",
"expires": "2026-01-21T00:00:00Z",
"revocable": true,
"ledger": {"tx": "0x...", "status": "active"}
}
```

D) Drift & Safety: from math to behaviors

1) $\Delta\Lambda^P$ / FEG → UI Logic

- **Inputs** (optional sensors + behavior): HRV trend, RR, skin temp delta, typing latency, pause requests, language volatility, self-tags (dizzy, overwhelmed).
- **Operator**: Drift risk increases when *HRV drops*, *latency spikes*, or *self-tags* rise; FEG anticipates collapse → **Offer Silence**.
- **Actions** (never coercive): Show **Sip** (default), or **Salt** if dizzy tag present; **Still** (60–120s breathe) if HRV low + agitation copy.
- **CSL limits**: cap to 1 suggestion / 10 min; switch to *quiet mode* if declined twice.

2) $\Xi 12$ Ethical Firewalls (health)

- **Pre-emptive**: soften/replace punitive diet terms; content notes for ED-phrases; purpose-gated outbound links.
- **Real-time**: interrupt with grounding copy; *escalate to human* (opt-in) only if user requests.
- **Post-session**: redactable notes; 24h auto-erase default; exports require fresh consent check.

E) Interop Recipes (ready for pilots)

- **Biosensor** → **FHIR**: transform ring JSON → `Observation` (LOINC HR/HRV/SpO₂/temp), `DiagnosticReport` for summaries; queue with 15–60s cadence.
- **Reflect** → **xAPI**: `actor=user`, `verb=answered`, `object=prompt:p2`, `result=tags:steady/need:rest`.
- **Groups** → **DIDs**: Microcircle IDs issued as DIDs; alignment check asserts purpose before any metric leaves device.
- **Coach Export**: `Bundle` of selected FHIR resources + a human-readable PDF; both stamped with consent ledger entry.

F) Accessibility & ND Support (production specifics)

- **Reading**: dyslexia-friendly font; 90-char line; adjustable spacing; motion-reduced mode.
 - **Input**: large-tap targets; TTS for prompts; STT for logs; haptic breathe cues.
 - **Sensory**: palette stays low-saturation; *Enter in Silence* launches text-off + breath overlay.
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G) MVP Scope & Sprints

Pilot-fit (90 days)

S0 – Foundation (Weeks 0–2) - Consent/de-ID flows; session auto-erase; CSL gate v0; feature store v0; Subscriptions/Webhooks smoke tests.

S1 – Experience & Telemetry (Weeks 3–8) - Ship: Sleep-101, Stress-201, Metabolism-101 *micro-lessons*; enable Prompt Orchestrator; PER/MLEM/TCI logging; privacy dashboard v1.

S2 – Inclusion & Pop Views (Weeks 9–13) - Add ASD EchoBraid protocol; population dashboards; first audit cycle; equity gap close target ≥ 0.15 .

Stretch to MVP-2 (4–8 months more)

- DNA-guided protocols; PIRTM stabilized operators; coach portals; expanded ring SDK features; ResearchStudy bindings.
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H) KPIs & Guardrails (no gamification)

- **Steadiness after pause** (% sessions reporting calmer/steadier).
 - **Opt-in to proactive pause; auto-delete rate** ($\geq$ default).
 - **One-protocol adherence** (self-rated, 1–6 weeks).
 - **High-drift episode rate** ($\downarrow$ across weeks); **overload** ≤ 0.10 ; **privacy pass-rate** ≥ 0.98 .
 - **p50 personalization** $\sim 100\text{ms}$; **event**→**feature p95** ≤ 5 min (on synced nodes).
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I) Acceptance Criteria (pilot)

- Users can complete a **meal log in** ≤ 3 taps with **no calories required**.
 - **Offer Silence** appears within **2s** when drift crosses threshold; declining twice silences prompts for **10 min**.
 - **One-tap wipe** fully erases local session text; exports demand a fresh consent confirmation sheetlet.
 - **Coach export** compiles only the user-selected metrics and redacted text; bundle is sealed with consent ledger.
 - **Offline mode**: full journaling and plan review work offline; sync resumes within 24h with conflict safety.
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J) Content & Copy (additions)

- **Overwhelm Note**: “Nothing to fix. We can pause.”
- **Salt Sheetlet**: “If lightheaded, a pinch of salt *with water* can help. If symptoms persist, consider a clinician.”

- **CSL Friendly Decline:** “Got it. I’ll quiet down for a bit.”
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K) Threat Model & Mitigations (focused)

- **Re-identification via shared reports** → Small-cell suppression; share-only aggregates by default; ledger traces viewable by user.
 - **Prompt overrun** → CSL gate + Reciprocity Balancer; backoff + quiet mode.
 - **Sovereign sync conflicts** → Last-writer-wins with user review queue; purpose-checks on re-share.
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L) Build Notes

- **Frontend:** React Native + local encrypted store; TTS/STT optional; responsive web shell for coaches.
 - **Pipelines:** ring→mobile→FHIR queue; feature store (energy steadiness, hydration sufficiency, meal rhythm, sleep regularity);
 - **Explainability:** every Omega Action shows *why* (tip-over inputs and top features) and a *decline* path.
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M) Next Artifacts (on request)

- Low-fi wireframes for *Session*, *Reflect*, *Coherence*, *Plan*.
- FHIR + xAPI mapping table (LOINC codes by metric).
- Consent/Ledger sequence diagram (offer, grant, revoke, export).